MALWARE FUNDAMENTALS

Quick Revision & Interview Preparation Notes

> ***One-Line Revision:*** *Malware is malicious software designed to steal data, damage systems, or gain unauthorized access to devices and networks.*

Definition

Malware stands for Malicious Software designed to:

- Damage systems
- Steal data
- Gain unauthorized access
- Disrupt operations

Main Goals of Malware

- Data theft
- System damage
- Credential stealing
- Financial gain
- Remote access

Common Types of Malware

Type	Description
Virus	Attaches to files and spreads when executed
Worm	Self-replicates across networks
Trojan	Disguised as legitimate software
Ransomware	Encrypts files and demands ransom
Spyware	Secretly monitors user activity
Adware	Displays unwanted advertisements
Rootkit	Hides malicious activities
Keylogger	Records keystrokes
Bot/Botnet	Infected device controlled remotely

Common Malware Delivery Methods

- Phishing emails
- Malicious attachments
- Fake software downloads
- Infected USB devices
- Drive-by downloads
- Exploiting vulnerabilities

Common Malware Indicators

- Slow system performance
- High CPU/network usage
- Unknown processes
- Disabled antivirus
- Pop-up ads
- Unauthorized login activity
- Suspicious outbound connections

Common IOCs in Malware Analysis

- Malicious IP addresses
- Suspicious domains

- File hashes
- Registry changes
- Abnormal processes
- Network connections

Important File Types & Risks

File Type	Risk
`.exe`	Executable malware
`.bat`	Malicious batch script
`.js`	JavaScript malware
`.vbs`	Visual Basic script malware
`.docm`	Macro-enabled malware
`.zip`	Hidden malicious payload

Malware Attack Flow

1. Malware delivered
2. User executes file/link
3. Malware installs on system
4. Persistence established
5. Data theft or system damage begins

Interview Questions & Answers

1. What is malware?

Malware is malicious software designed to damage, disrupt, or gain unauthorized access to systems.

2. Difference between virus and worm?

Virus	Worm
Needs user action to execute/spread.	Self-replicates automatically.
Attaches to host files.	Spreads through networks standalone.

3. What is a Trojan?

A Trojan is malware disguised as legitimate software to trick users into installing it.

4. What is ransomware?

Ransomware encrypts files and demands payment to restore access.

5. What are common signs of malware infection?

Slow system, high CPU usage, unknown processes, suspicious network traffic, and disabled security tools.

6. What is a file hash?

A unique cryptographic value used to identify and verify the integrity of files.

7. What are IOCs in malware analysis?

Indicators of Compromise (IOCs) are artifacts like malicious IPs, domains, file hashes, or suspicious behaviors that show a system has been compromised.

8. What is persistence in malware?

Techniques used by malware to ensure it remains active and automatically runs even after a system reboot.